TED (21) – 5092 REVISION 2021

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

QUALITY ASSURANCE AND TESTING

MODEL QUESTION PAPER – SET-1

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Sl no.	Question	Module Outcome	Cognitive level
1	List four thermal properties of polymers.	M 1.04	Remembering
2	Name two national standardsorganizations.	M1.01	Remembering
3	What is the significance of PRI in the quality control of rubbers?	M2.01	Understanding
4	Ammonium sulphate is used for the coagulation of latex in VFA number determination. Give reason.	M2.03	Understanding
5	Define M300 and how it is expressed.	M3.01	Remembering
6	Define abrasion loss.	M3.03	Understanding
7	State two tests for Water tanks.	M4.04	Understanding
8	Define indentation hardness index.	M4.03	Remembering
9	What is the significance of determining HDT of polymers?	M1.04	Understanding

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Describe purity test for MBT.	M1.03	Understanding
2	Describe the experiment to determine M.P by Kofler	M1.04	Understanding

	method.		
3	Explain the procedure to determine Viscosity of latex using Ostwald viscometer.	M2.03	Understanding
4	Define KOH number. Explain its significance.	M2.03	Understanding
5	Distinguish tear strength and tensile strength.	M3.01	Understanding
6	Explain the procedure to determine compression set of rubbers.	M3.02	Understanding
7	Explain the Quick burst strength test for PVC pipes.	M4.04	Understanding
8	Outline the specification tests for V-Strap.	M4.02	Understanding
9	Identify the procedure to determine TSC of latex.	M2.03	Understanding
10	Draw an MFI tester and label its parts.	M3.04	Understanding

PART C

III. Answer all questions from the following (6x 7 = 42 Marks)

		Module Outcome	Cognitive level
1	Explain the test procedure to determine brittleness temperature of polymers.	M1.04	Understanding
OR			
2	Outline the test method to determine flammability of self-supporting plastics.	M1.04	Understanding
3	Describe the significance and method to determine MST of concentrated latex.	M2.03	Understanding
OR			
4	Describe the significance and procedure to determine Mg content of NR latex.	M2.03	Understanding
5	Describe the significance and procedure to determine tension set of rubbers.	M3.02	Understanding
OR			
6	Define Hardness and explain the procedure to determine it.	M3.03	Understanding
7	Describe three specification tests for surgical gloves.	M4.03	Understanding
OR			

8	Explain izod and charpy impact test for water tanks.	M4.04	Understanding
9	Describe the basic set up to determine VST of polymers.	M1.04	Understanding
OR			
10	Explain Fisher john's method to determine the M.P of polymers.	M1.04	Understanding
11	Explain the specification test for examination gloves.	M4.03	Understanding
OR			
12	Explain two specification tests for M.C sole.	M4.02	Understanding

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MODEL QUESTION PAPER – SET-2

Time: 3 hours Maximum Marks: 75

PART A

IV. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Sl no.	Question	Module Outcome	Cognitive level
1	Define Dielectric strength.	M 1.04	Remembering
2	List two thermal properties of polymers.	M1.04	Understanding
3	Differentiate VFA number from KOH number.	M2.03	Understanding
4	Calculate protein content of NR latex if nitrogen content is 0.6.	M2.01	Understanding
5	Write the Unit of Tensile strength.	M3.01	Remembering
6	State two uses of MFI tester.	M3.04	Understanding
7	Distinguish Charpy and izod test.	M4.04	Understanding
8	List the T.S and EB of Surgical gloves.	M4.03	Remembering
9	Define VST.	M1.04	Remembering

PART B

V. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Discuss the role of ASTM in standardization and quality assurance.	M1.02	Understanding
2	Identify the procedure to determine percentage purity of	M1.03	Understanding

	DPG.		
3	Explain the significance of MST test of NR latex.	M2.03	Understanding
4	Explain the significance of doing Nitrogen content of ISNR.	M2.01	Understanding
5	Describe the compression set apparatus used for testing rubbers.	M3.02	Understanding
6	Draw the diagram of Dunlop tripsometer and label the parts.	M3.03	Understanding
7	Illustrate split tear strength for rubber hawai chappal.	M4.02	Understanding
8	Explain autoclave test for surgical gloves.	M4.03	Understanding
9	Identify the procedure to determine DRC of latex.	M2.01	Understanding
10	Explain the parts of a tensile testing machine.	M3.01	Understanding

PART C

VI. Answer all questions from the following (6x 7 = 42 Marks)

		Module Outcome	Cognitive level
1	Explain the test procedure to determine arc resistance of polymers. Mention its significance.	M1.04	Understanding
	OR		
2	Define LOI and explain the procedure to determine LOI of polymers.	M1.04	Understanding
3	Describe the method to determine MST of concentrated latex.	M2.03	Understanding
	OR		
4	Classify latex based on its alkali content. Describe the procedure to determine alkalinity of Latex.	M2.03	Understanding
5	Define resilience and explain the procedure to determine it.	M3.03	Understanding
	OR		
6	Define tension set and Explain the procedure to determine it.	M3.02	Understanding
7	Describe the joint adhesion strength and leakage test of cycle rubber tube.	M4.02	Understanding

OR			
8	Describe the method to determine shrinkage and Hardness of Rubber hawai chappal.	M4.02	Understanding
9	Describe guarded hot plate method to determine thermal conductivity of polymers.	M1.04	Understanding
OR			
10	Define HDT and explain the apparatus or basic set up to determine HDT.	M1.04	Understanding
11	Define Hardness and explain the parts of a Shore A durometer	M3.03	Understanding
OR			
12	Explain the significance principle and method to determine crack initiation resistance of rubbers using Demattia flex tester.	M3.03	Understanding